

Who wins Col on a $3 \times n$ board?

A friendly guide to a proof that the second player always wins

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The one-sentence answer

On an empty board with 3 rows and any number of columns, the player who moves *second* can always win Col, no matter how cleverly the first player plays.

This guide explains why, step by step, with lots of pictures. You do not need any background beyond knowing how to play a board game. A more formal version of the same argument is written for mathematicians in the companion paper.

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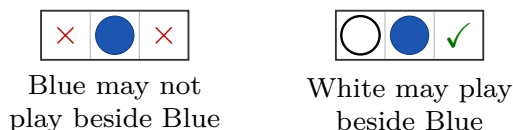
1 Meet the game

Col is a game for two players, whom we call **Blue** and **White**. They take turns placing a stone of their own colour on an empty square of a grid. There is exactly one rule:

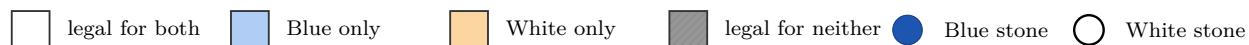
The only rule

You may **not** put a stone right next to (above, below, left or right of) one of **your own** stones. Next to your opponent's stones is fine.

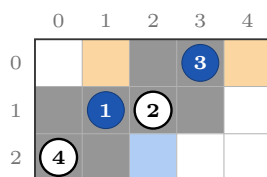
If it is your turn and you have no legal square, you **lose**. There are no draws, and there is no luck.



As the game goes on, some squares become off-limits to one player, some to the other, and some to both. We colour each empty square to show who may still play there:



Here is a position on a board with 3 rows and 5 columns after four moves. The numbers show the order the stones were played:

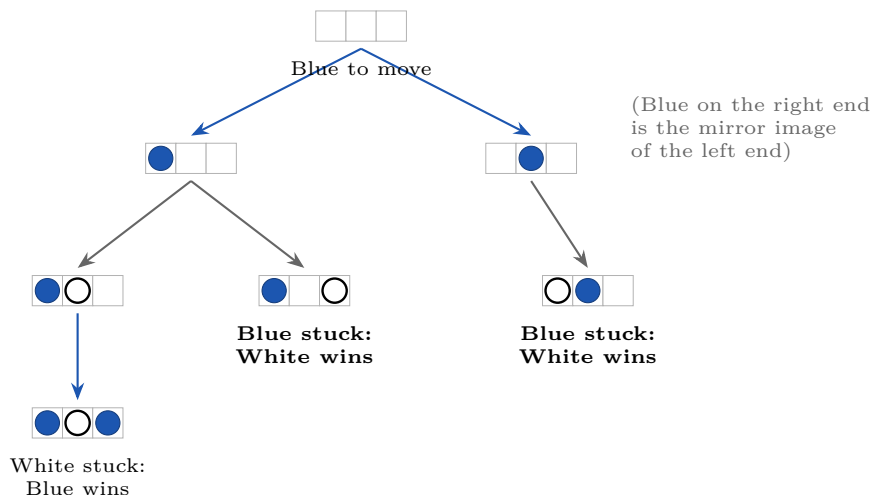


Blue's stone 1 makes the squares around it pale orange: only White may still play there. White's stone 2 makes its neighbours pale blue. The grey squares are dead for both players. A player's legal squares are the white ones plus those tinted in her own colour, so this picture shows at a glance what each player can still do.

2 What does “the second player wins” mean?

Game theorists say a game has **value zero** if whoever moves second can always win. It does not matter whether Blue or White starts; the one who waits wins.

The smallest example is a strip of three squares. Here is every way the game can go when Blue starts (the right end is a mirror image of the left, so we skip it):



If Blue starts in the middle, White plays at an end and Blue is stuck, since the other end is next to Blue's stone. If Blue starts at an end, White should take the *other* end. Then the middle is next to Blue's stone, and Blue is stuck again. Either way, White wins. The same happens if White starts, with the colours swapped. So this little strip has value zero.

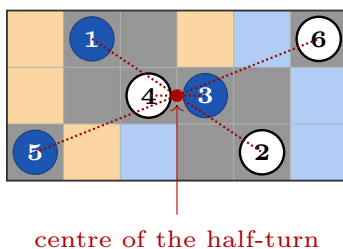
Our goal

Show that the empty board with 3 rows and n columns (call it the $3 \times n$ board) has value zero, for *every* n : 1, 2, 3, ..., 1000, and forever.

We can never try every n , and a single board can have more possible games than there are atoms in the universe. We need ideas.

3 Even widths: the copycat trick

Suppose the board has an even number of columns, say 6. Imagine spinning the board half a turn around its centre. Every square lands on a *different* square. The second player simply copies every move to the spun-around square:

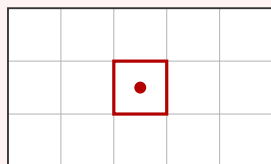


Why is the copy always legal? The board always looks the same after the spin, with the colours

swapped. If Blue's move was legal, then White's spun copy is legal for exactly the same reason. So the copycat never runs out of moves, and the other player gets stuck first.

Why this fails for odd widths

With an odd number of columns, the centre square spins onto *itself*. If the first player takes the centre, there is nothing to copy onto.



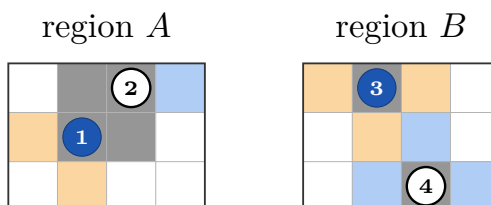
the centre cell is its own mirror image

The rest of this guide is about odd widths.

4 Two big ideas

4.1 Big idea 1: games can be added

If a board splits into separate regions that do not touch, you can think of it as several games played at once: on each turn you choose one region and move there. Game theorists call this a *sum* of games.



White always answers in the same region as Blue.
If White wins each region as second player, she wins the sum.

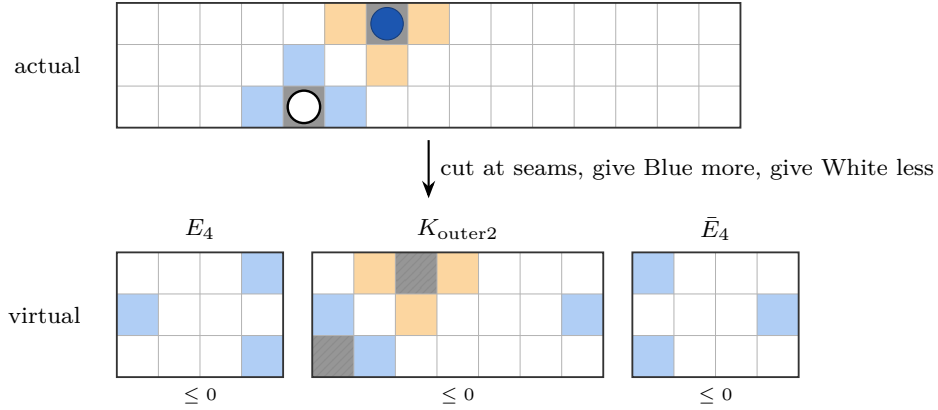
Adding second-player wins

If White wins each region when Blue moves first there, White also wins the whole sum. Her strategy: always answer in the same region Blue just played in.

4.2 Big idea 2: cheating in Blue's favour

Our boards do not split into separate regions by themselves. So we pretend they do. We cut the board into pieces and change the rules a little, always in **Blue's** favour:

- Blue may keep every square she really has, and we may give her extra ones;
- White may only use squares she really has, and we may take some away;
- at each cut, White must not be allowed to play on both sides of the cut in the same row.



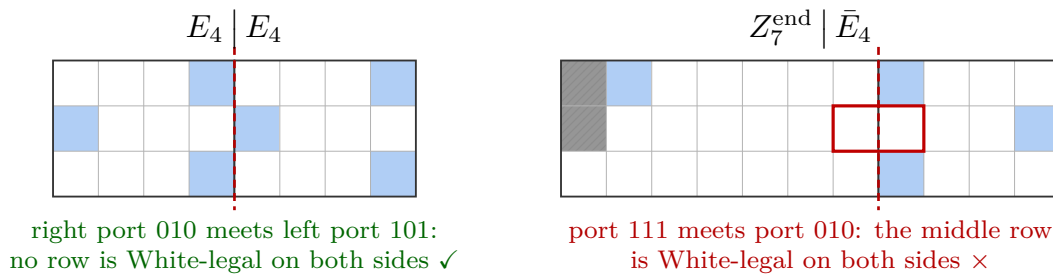
If White can still win this “rigged” game, she can certainly win the real one. Her real moves copy her virtual moves, and Blue’s real moves are also available in the virtual game. This is the **comparison principle**, the main tool of the whole proof.

Why the third rule matters

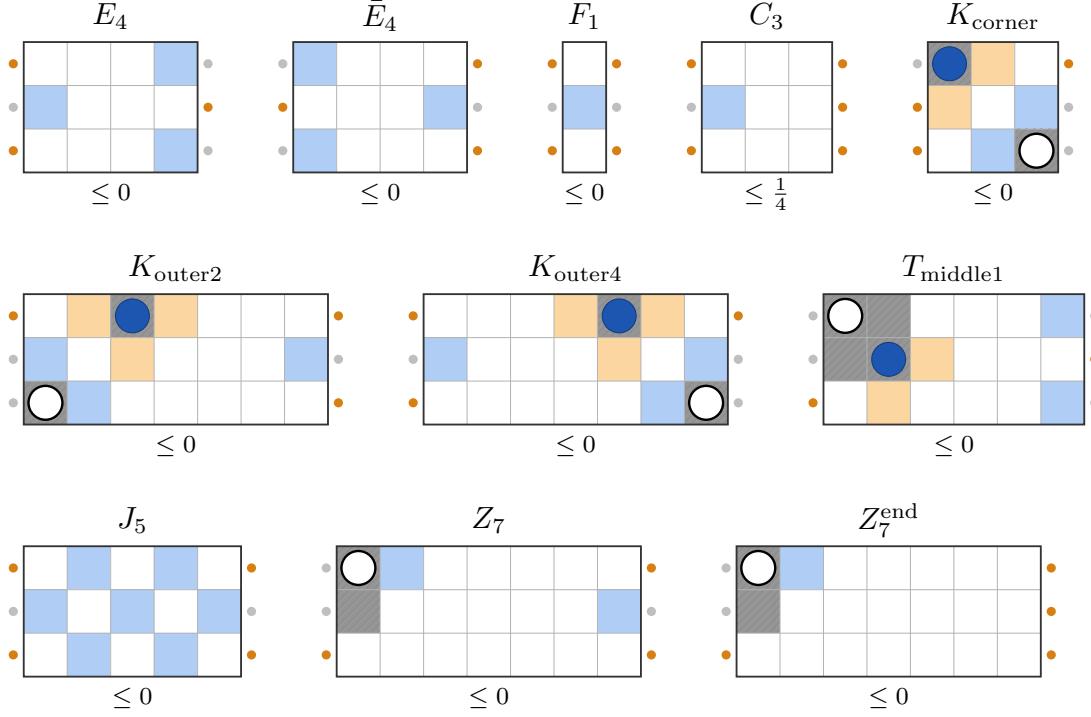
In the real game, two squares side by side across a cut really are next to each other. If White could play on both of them in the rigged game, she would be allowed something the real rules forbid. So at every cut we check that each row is White-legal on at most one side.

4.3 Pieces and plugs

The pieces we cut the board into are called **gadgets**. Each gadget is a small 3-row board with some squares already restricted. The pattern of White-legal squares on a gadget’s left and right edges works like a *plug*. Two gadgets may sit side by side only if their plugs never both have White in the same row:



Here is the whole toolbox. Under each gadget is a guarantee that a computer has checked. “ ≤ 0 ” means White wins that gadget when Blue moves first there.

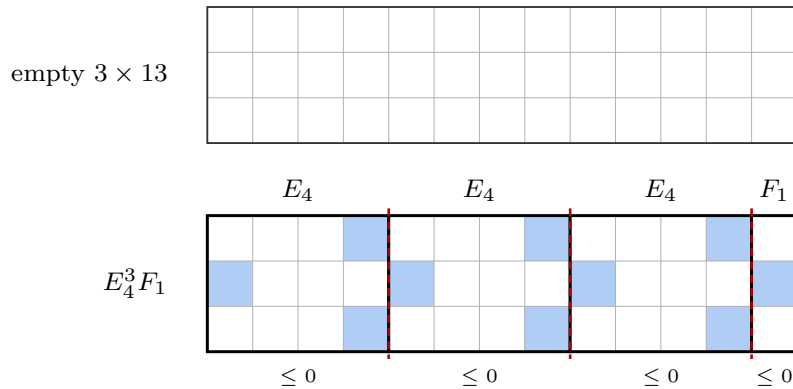


4.4 Scores that are not whole numbers

Some gadgets come with a score like -1 or $\frac{1}{4}$ instead of 0. Think of it as a count of spare moves: -1 means White is a whole spare move ahead, and $+\frac{1}{4}$ means Blue is a quarter of a move ahead. (Fractional moves really do make sense in game theory!) When we add regions we add their scores. If the total is below zero, White wins even if it is her turn to move.

5 Widths 5, 9, 13, 17, ... (one more than a multiple of 4)

For these widths the pieces fit like a brick wall: repeat the four-column gadget E_4 and finish with the one-column gadget F_1 .



Every plug fits, and every piece scores ≤ 0 . So White wins when Blue starts. By the colour swap, Blue wins when White starts. So the board has value zero.

The colour-swap shortcut

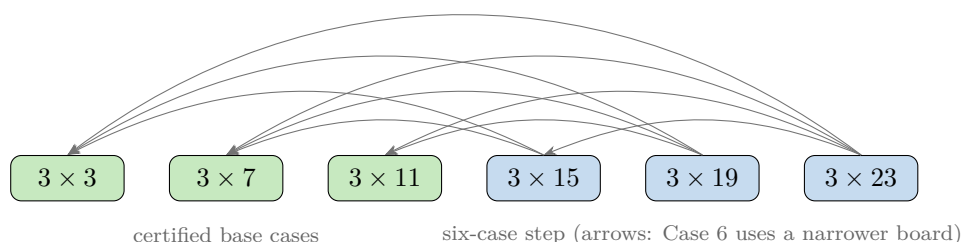
The empty board looks identical if you swap the names Blue and White. So if the second player wins when Blue starts, the second player also wins when White starts. We only ever need to check Blue starting.

6 Widths 3, 7, 11, 15, ... (three more than a multiple of 4)

This is the hardest family, and it needs a new idea: **induction**, the domino effect. We show:

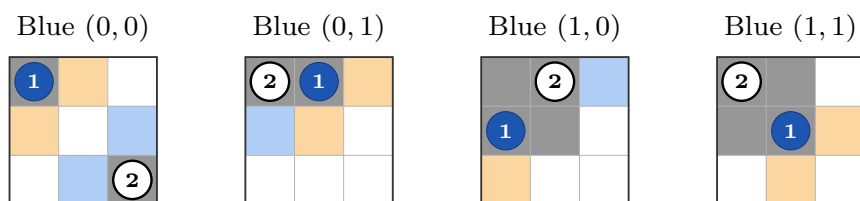
1. the boards of width 3, 7 and 11 have value zero (checked by computer);
2. if every narrower board in the family has value zero, then so does the next one.

Then width 15 follows from 3, 7 and 11; width 19 from those and 15; and so on forever, like falling dominoes.



6.1 The starting dominoes

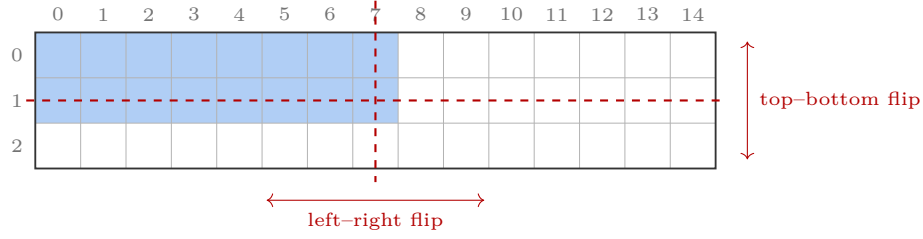
For the 3×3 board there are only four genuinely different first moves for Blue. Here is White's winning answer to each:



Widths 7 and 11 are handled the same way, with 8 and 12 first moves, and a computer checks every answer.

6.2 Only a quarter of the board matters

Flipping the board top-to-bottom or left-to-right does not change the game. So we only need to answer Blue's first move in the shaded part:



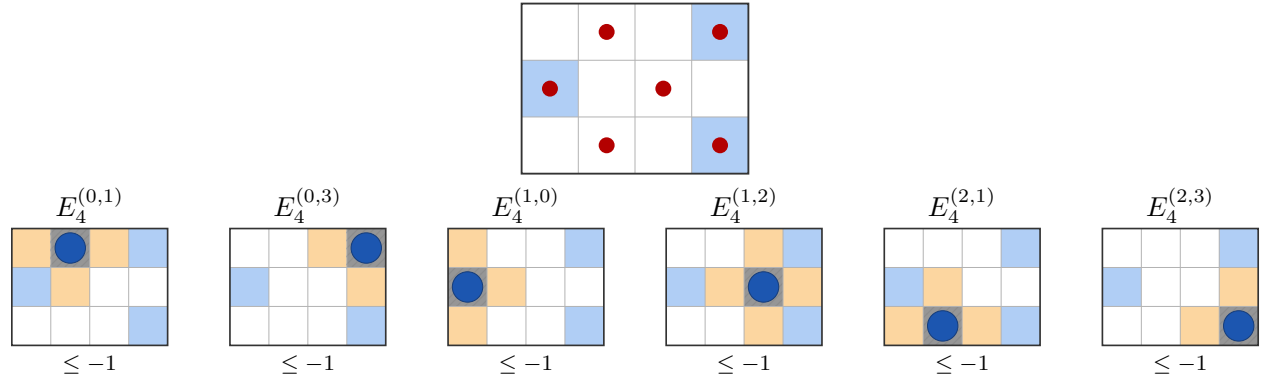
We sort those first moves into six cases, shown here for width 23:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
0	2	1	3	1	4	1	3	1	4	1	3	1											
1	1	5	1	6	1	5	1	6	1	5	1	6											
2																							

6.3 The six cases, in pictures

In each picture the top board is the real game after Blue's first move (and White's answer, if there is one). The bottom board is the rigged version, cut into gadgets. Circles show where the stones went.

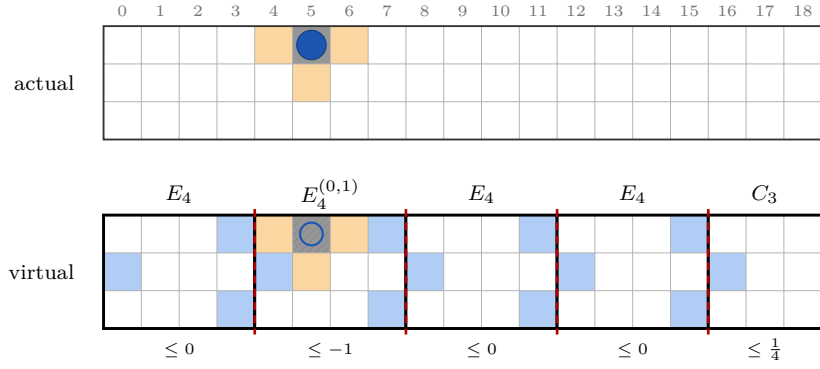
Case 1: Blue plays on an “odd” square. Colour the board like a chessboard; “odd” squares are those where row plus column is odd. Blue's move lands inside one of the E_4 bricks and damages it so badly that it scores -1 : White is a whole move ahead.



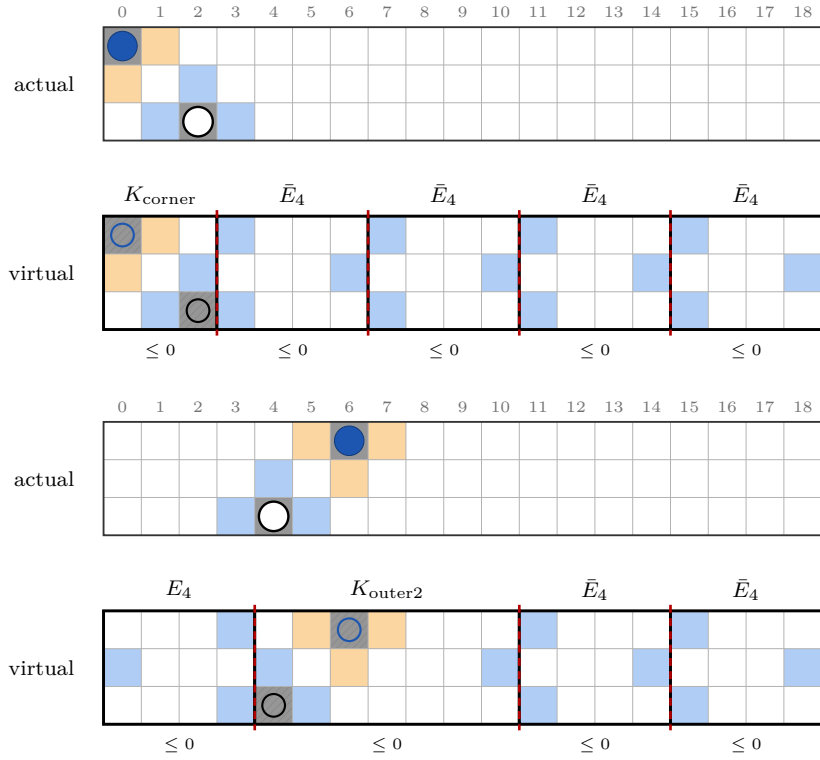
The end-cap C_3 gives Blue back only a quarter:

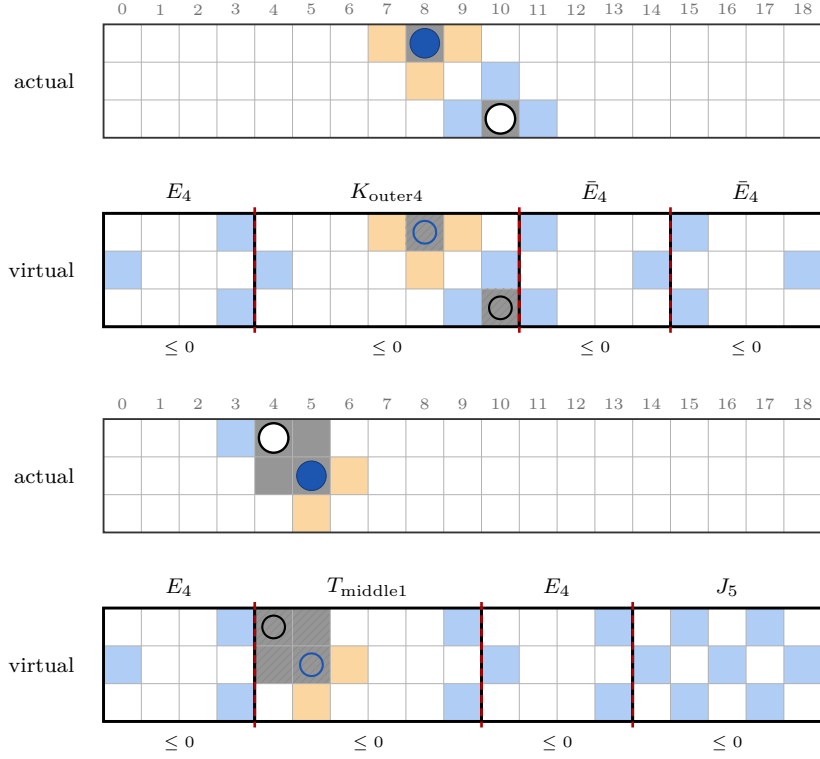


The total is $-\frac{3}{4}$, below zero, so White wins even though it is her turn. She does not need a specific answer at all!

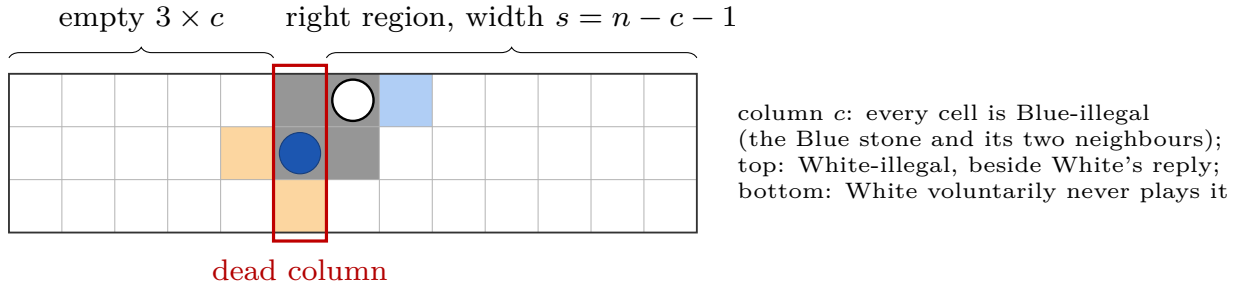


Cases 2 to 5: special answers. When Blue plays an “even” square in the shaded area, White makes one specific answer, and a special gadget around the two stones scores ≤ 0 .

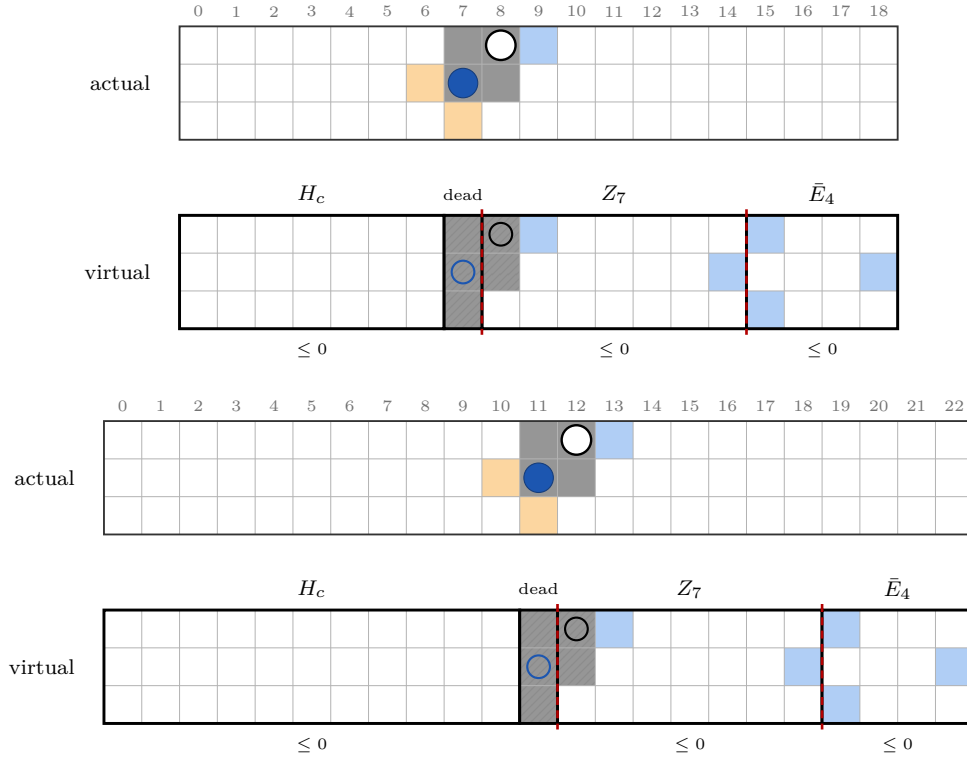




Case 6: the clever one. Blue plays in the middle row, in a column three more than a multiple of 4. White answers diagonally up and to the right. Now look at Blue's column:



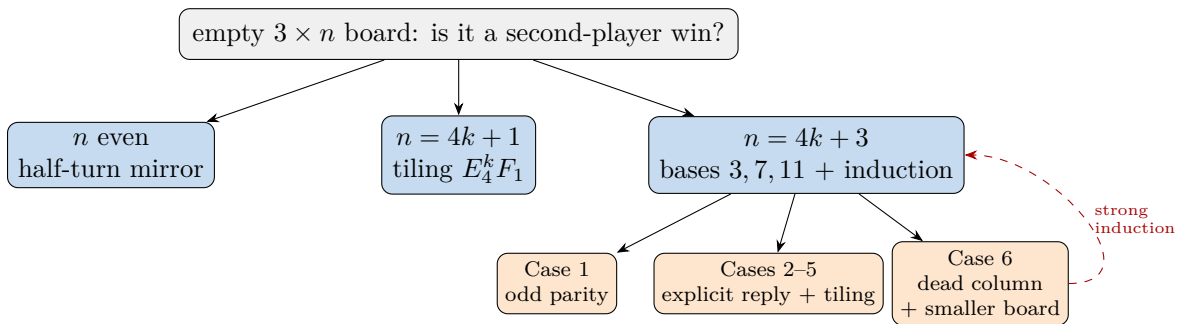
Blue can never play in that column again: one square has her stone, and the other two are next to it. White can't use the top square, which is beside her own stone, and she *promises* never to use the bottom one. The column has become a wall. To its left is an ordinary empty board, narrower than the one we started with and in the same family, and we already know it has value zero (the domino before). To its right sit certified gadgets.



Why this is the heart of the proof

Case 6 is the only place the dominoes are used. One stone plus one answer turns a big board into a smaller copy of the same problem. That is exactly what induction needs.

7 The whole proof on one page



- **Even width:** copy every move through the centre.
- **Width $4k + 1$:** a wall of E_4 bricks capped by F_1 .
- **Width $4k + 3$:** three starting dominoes, then six cases for Blue's first move. Case 1 wins on the score, Cases 2–5 use a special gadget, and Case 6 builds a wall that leaves a smaller board.

8 What did the computer do, and what did people do?

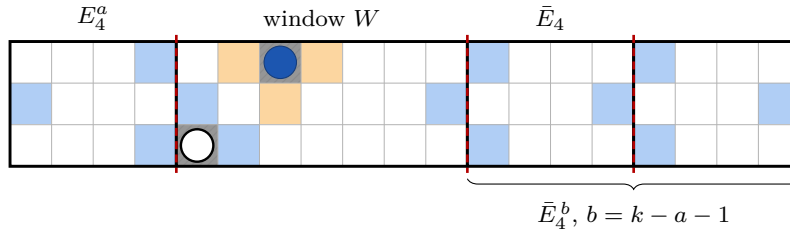
Each gadget's guarantee concerns a small, finite board, so a computer can check it completely. For each gadget the stored certificate lists, for every Blue move, White's answer, and a short program replays all of these. Every certificate is checked in a few seconds, without any searching.

The unlimited part, “for every width”, cannot be checked by trying widths one at a time. It rests on arguments: the copycat trick, the comparison principle, the colour swap, the plug table, and the dominoes.

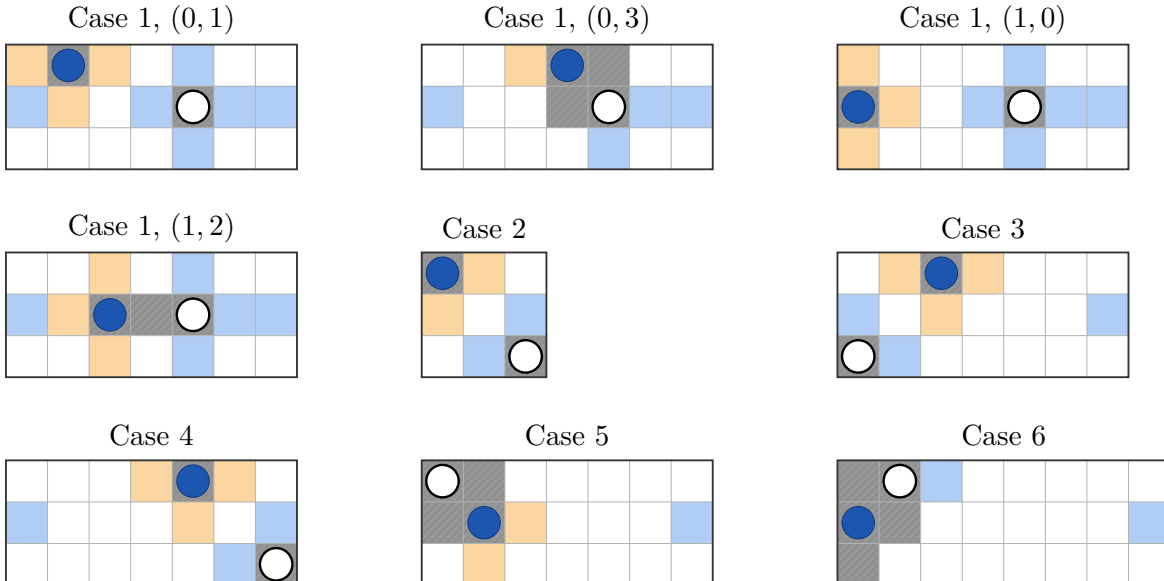
A computer checked the whole argument too

All of those arguments have also been written in **Lean**, a *proof assistant*: a program that accepts a proof only if every single step follows from the rules of logic. Lean checked the complete statement “for every width n , the player who moves first on the empty $3 \times n$ board loses”, stated with the real rules of Col, stones and all. It checked the gadget certificates as well. Nothing is left to trust except Lean itself.

The Lean version uses an even simpler pattern for every case: bricks, then one small *window* around Blue's first move and White's answer, then bricks again. The window is just the real board near the two stones, with White giving up the few squares her plugs require.



Here are all nine windows it uses:



With these windows the domino step already works at width 11, so the starting dominoes are just the 3×3 and 3×7 boards.

What is still unknown

Nobody yet knows whether the second player always wins on boards with 5 rows and an odd number of columns, or on odd-by-odd boards in general. The wall trick of Case 6 does not work there, because one stone does not block a whole column of 5 squares.

Glossary

Value zero The second player can always win.

Gadget A small certified piece of board with a score guarantee.

Plug / port The White-legal pattern on a gadget's left or right edge.

Comparison principle If White wins a version rigged in Blue's favour, she wins the real game.

Induction Proving every case by proving the first ones and showing each case follows from earlier ones.

Certificate A stored list of White's answers that a computer can replay to confirm a gadget's guarantee.